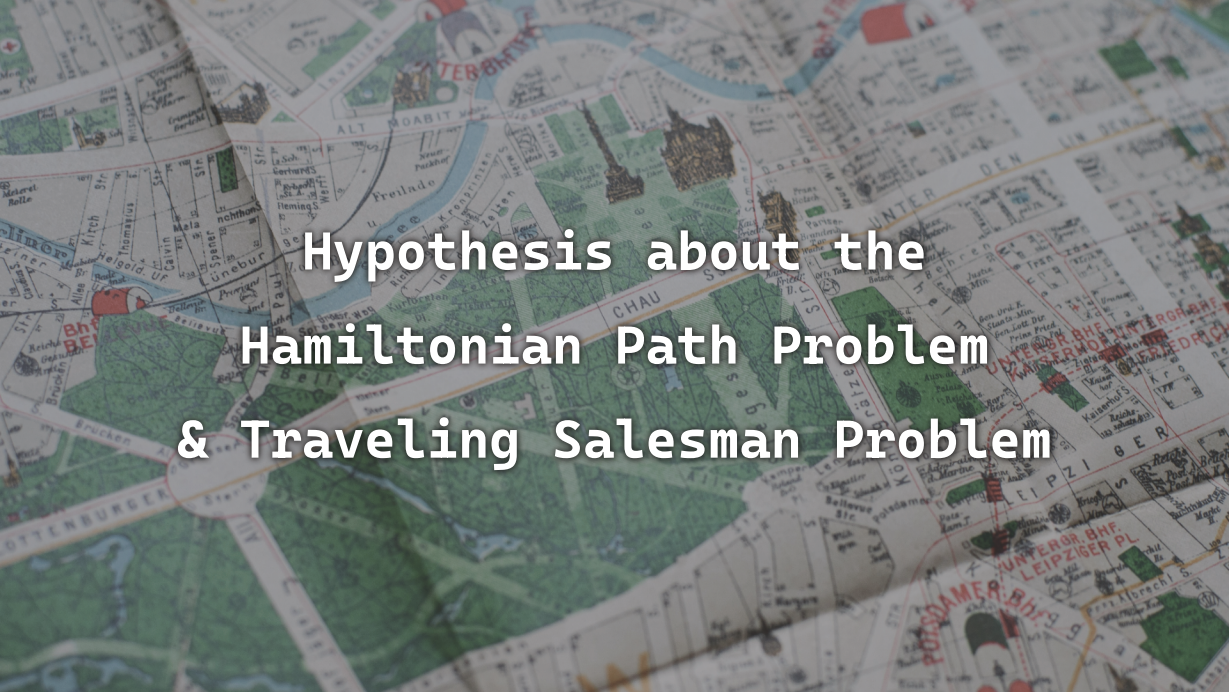
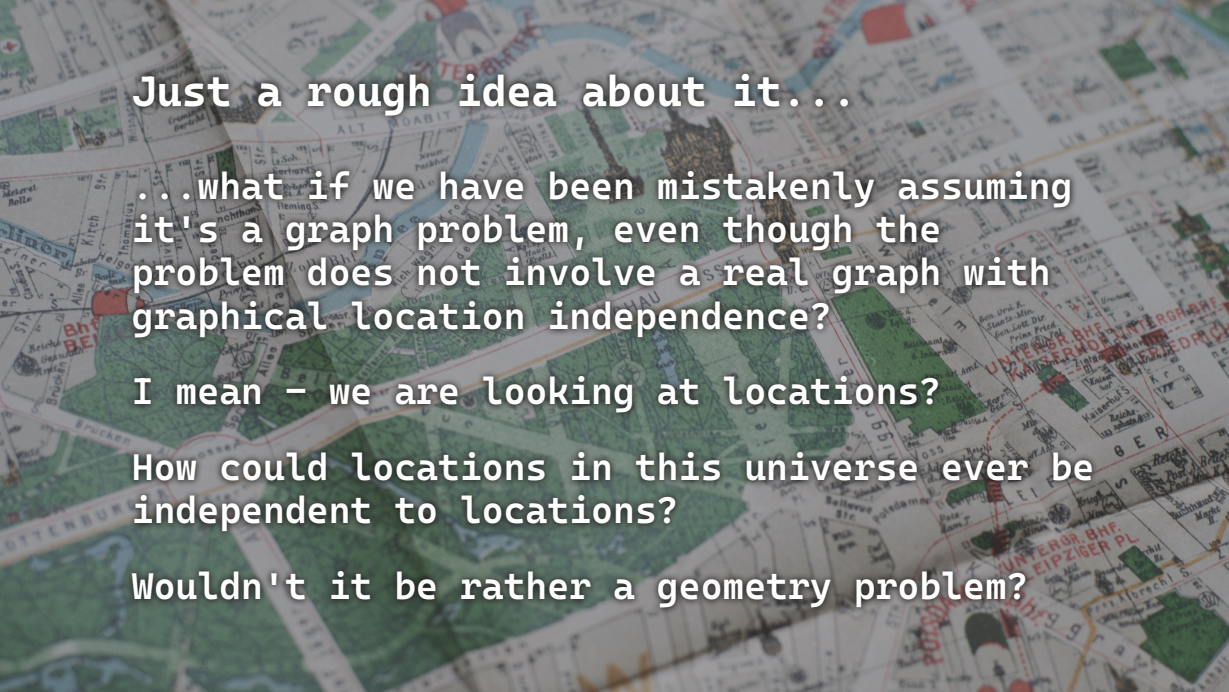




Hypothesis about the Hamiltonian Path Problem & Traveling Salesman Problem



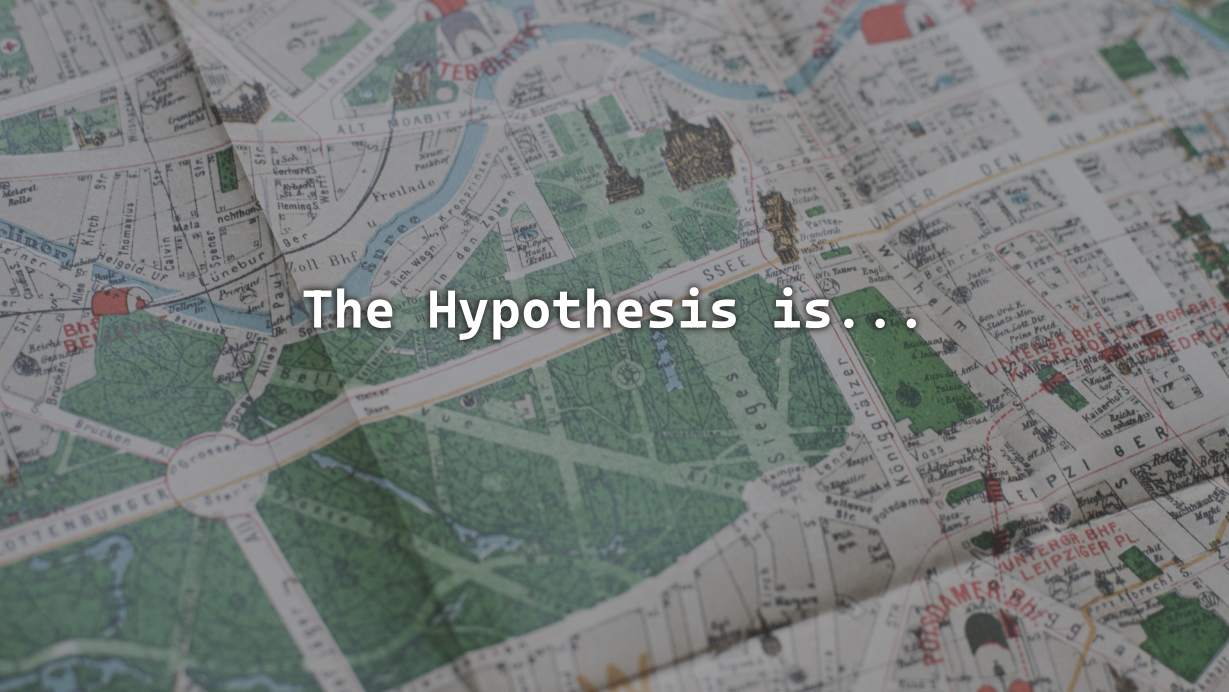
Just a rough idea about it...

...what if we have been mistakenly assuming it's a graph problem, even though the problem does not involve a real graph with graphical location independence?

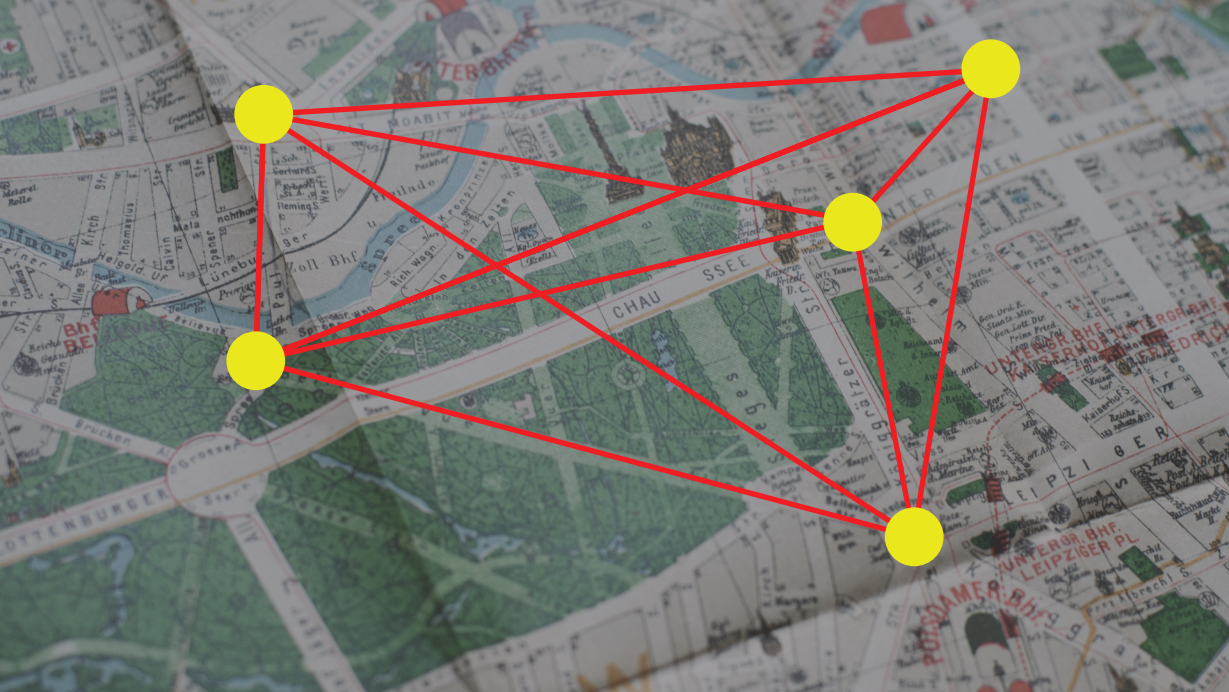
I mean - we are looking at locations?

How could locations in this universe ever be independent to locations?

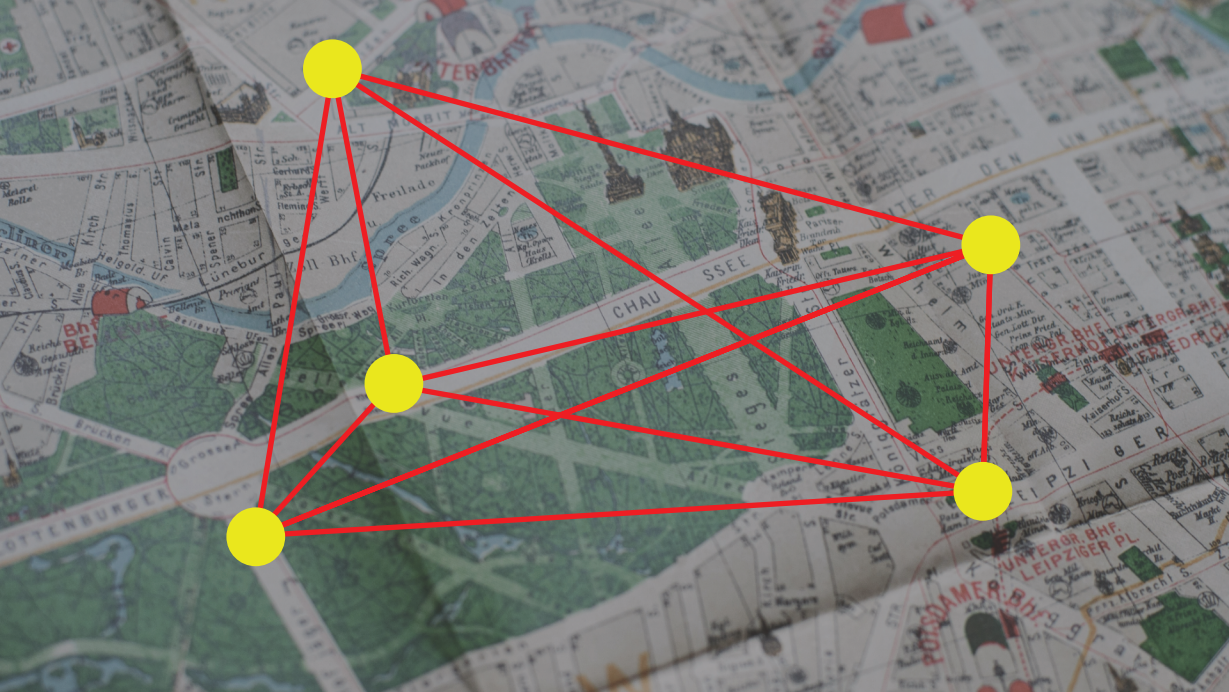
Wouldn't it be rather a geometry problem?



The Hypothesis is...



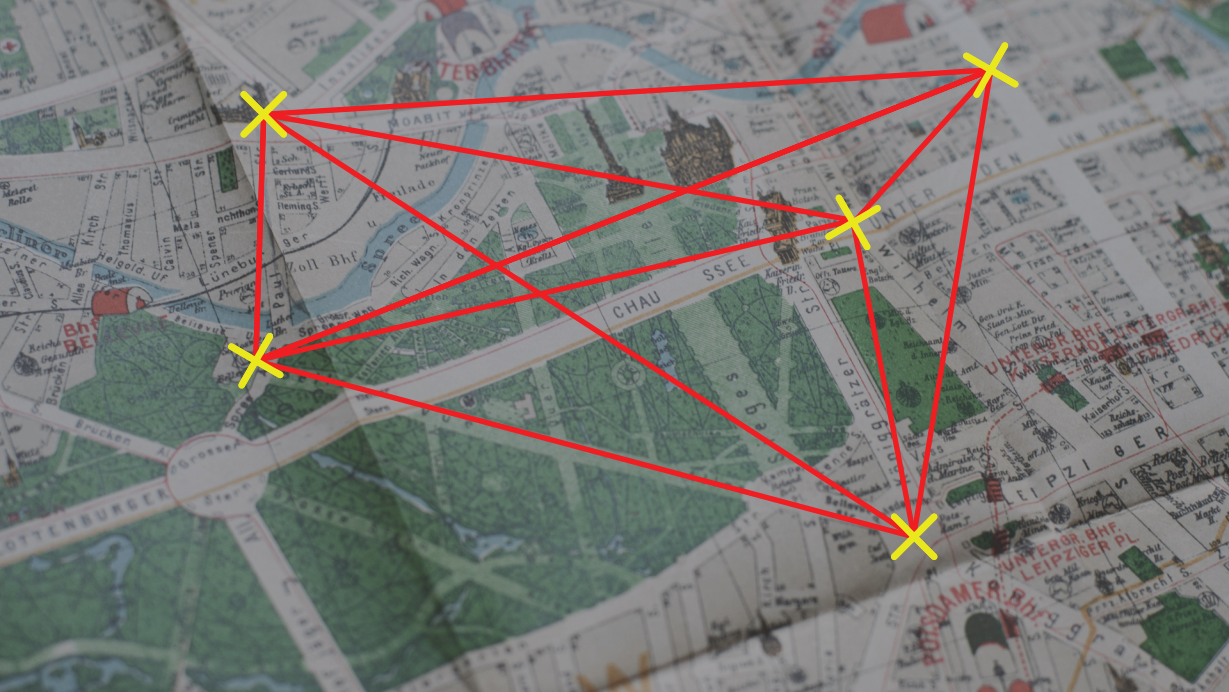


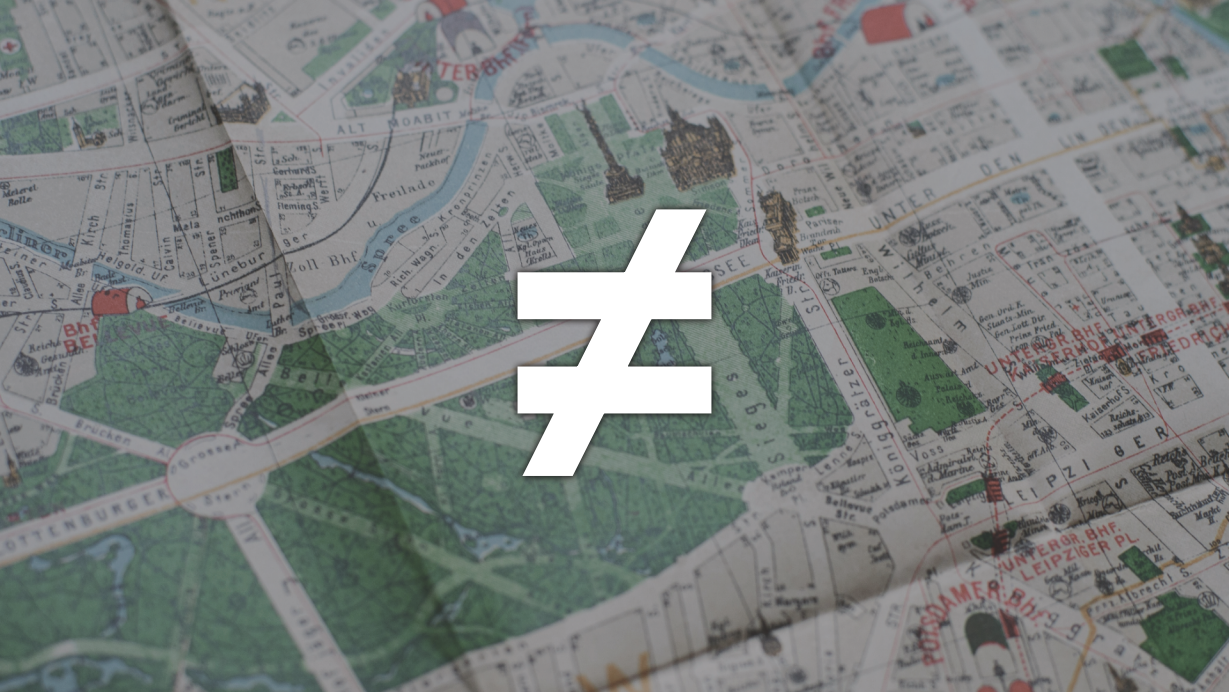


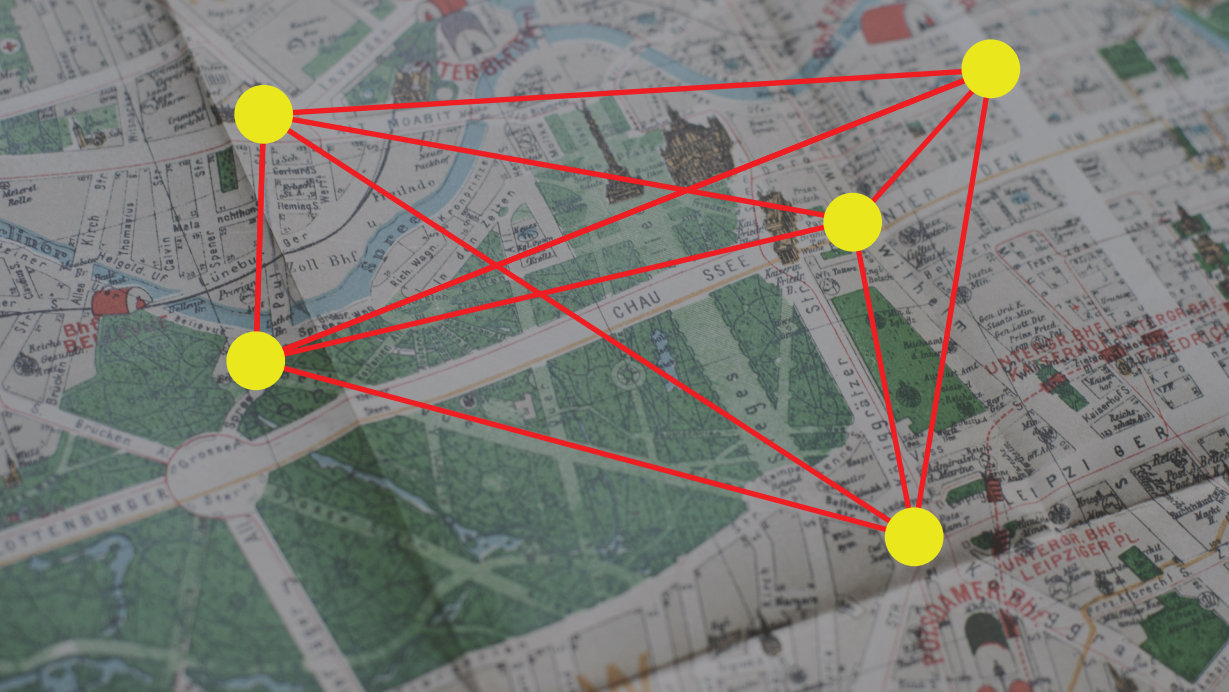


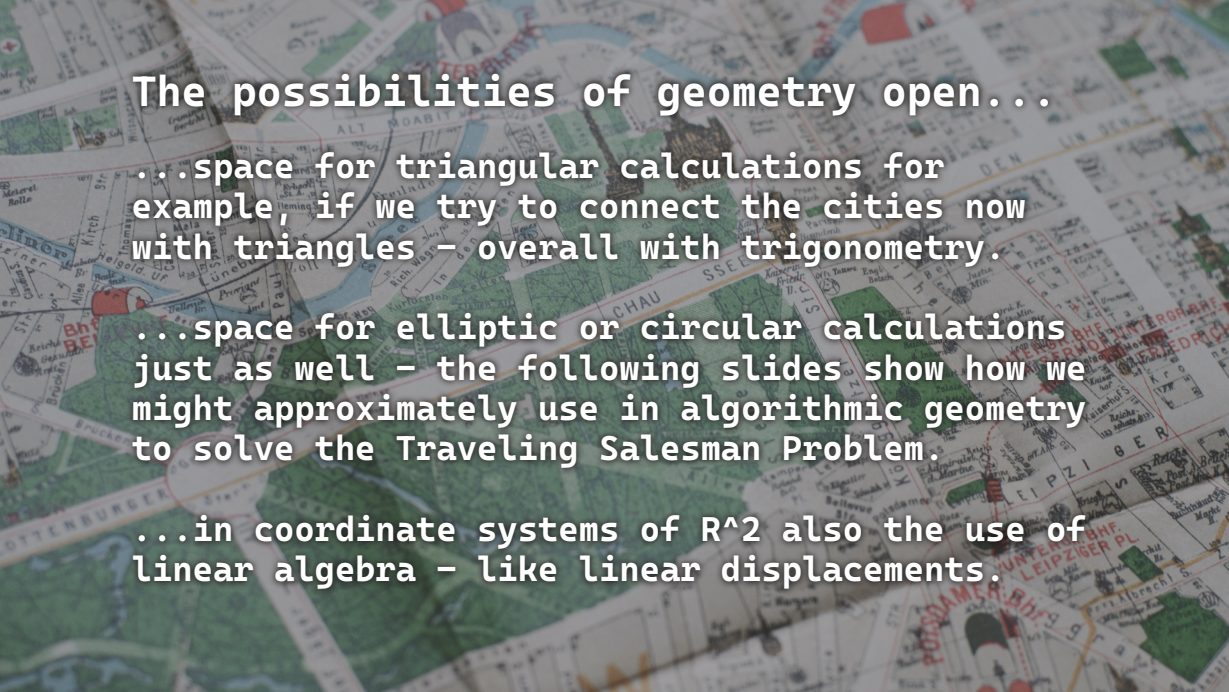










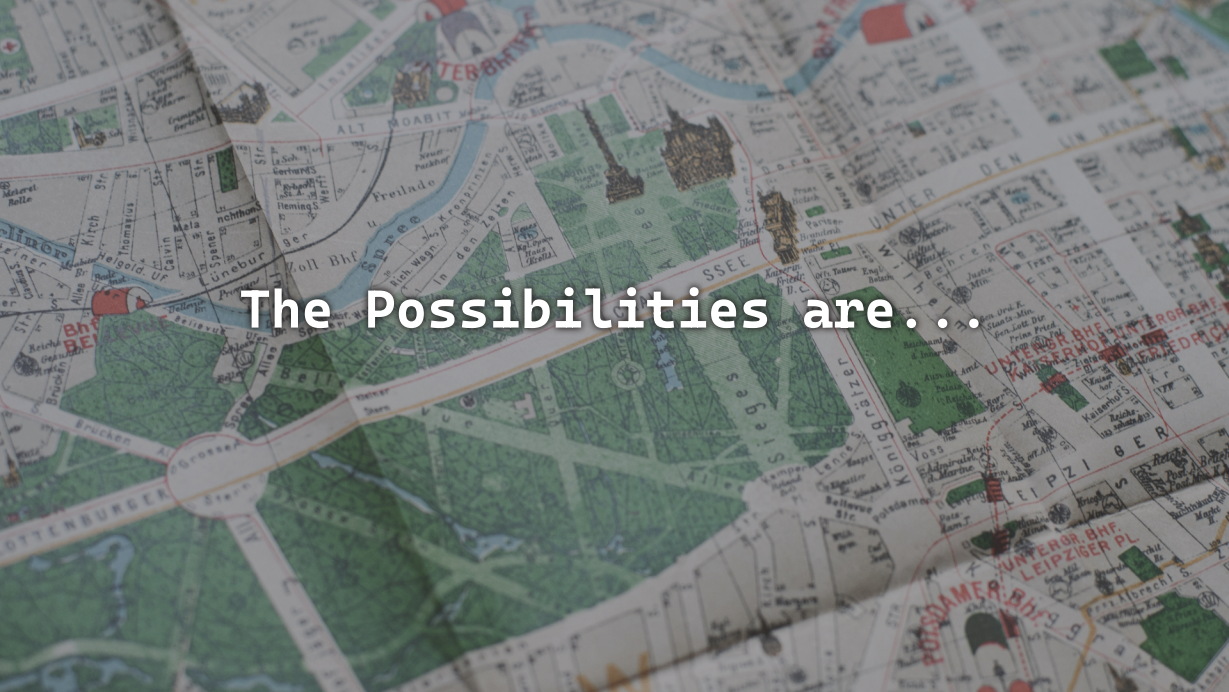


The possibilities of geometry open...

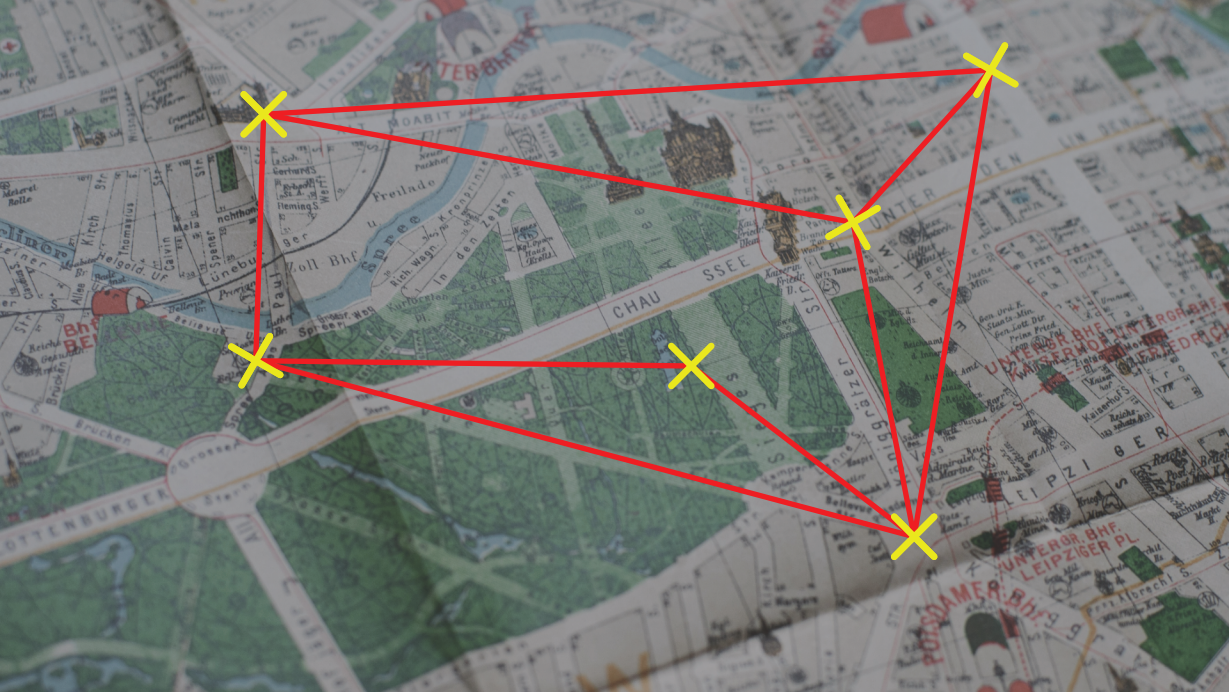
...space for triangular calculations for example, if we try to connect the cities now with triangles – overall with trigonometry.

...space for elliptic or circular calculations just as well – the following slides show how we might approximately use in algorithmic geometry to solve the Traveling Salesman Problem.

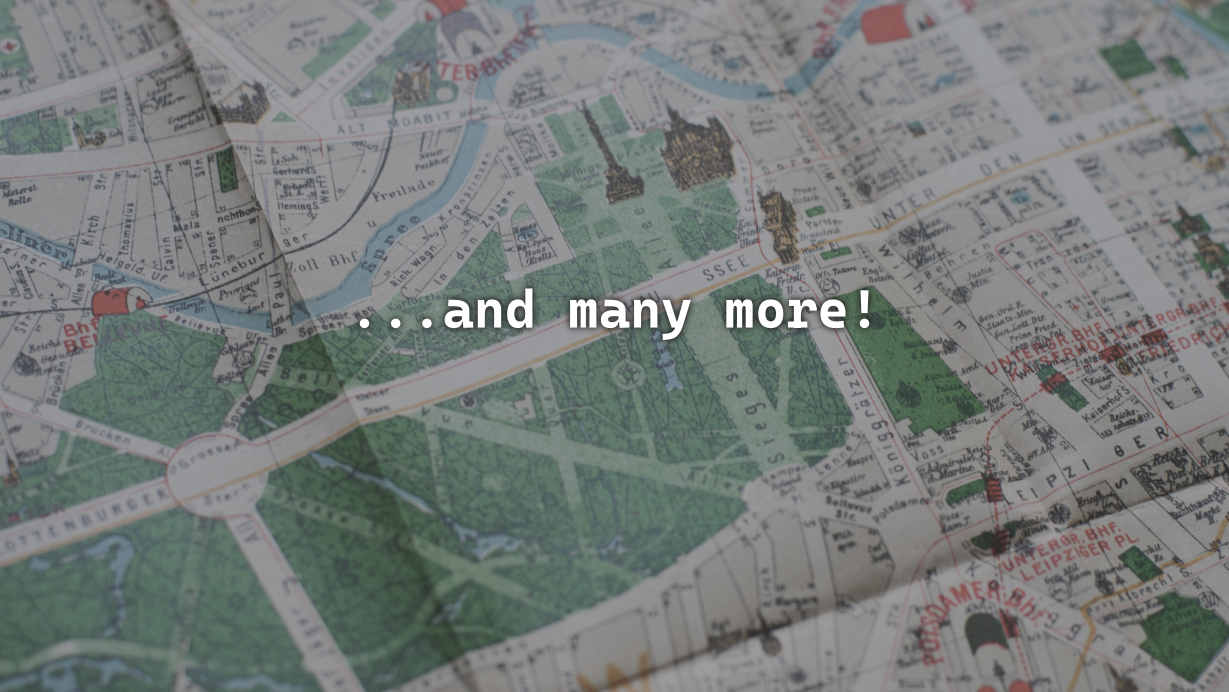
...in coordinate systems of R^2 also the use of linear algebra – like linear displacements.



The Possibilities are...







...and many more!



Idea & Presentation by Marc-Michel Münch

12th December 2025